



Network Traffic Forensics Report

Incident Title: Botnet / DoS Analysis
Incident Date: 18 August 2011 (01:39:35 – 01:53:24)
Classification: Confidential / Internal Use Only
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Executive Summary

This report presents a forensic analysis of network traffic captured over a 13.8-minute window. Using Python for data processing and Power BI for visualization, the investigation identified abnormal traffic patterns, including a massive spike originating from distributed source IPs targeting a single victim.

Key Finding: The attack is consistent with a coordinated, multi-protocol DoS/Botnet attack, utilizing multiple source IPs to flood the victim infrastructure.

Impact: The observed traffic would have caused significant bandwidth saturation, high latency, and likely service unavailability for legitimate users during the attack period.



Dataset Description

Attribute	Details
File Name	capture20110818-2.truncated.pcap
Data Type	Network Packet Capture (PCAP)
Capture Window	13.8 minutes
Peak Traffic	28 Million packets
Protocols	TCP, UDP, ICMP

Methodology

The analysis followed a structured forensic workflow:

1. **Data Extraction:** Converted raw PCAP files into a structured dataset using Python (Pandas).
2. **Sanitization:** Converted timestamps from Unix format to human-readable datetime and sorted chronologically.
3. **Aggregation:** Aggregated traffic statistics to identify peaks, anomalies, and top talkers.
4. **Visualization:** Created Power BI dashboards to visualize traffic spikes, protocol distribution, and targeting patterns.

Analysis & Findings

Traffic Volume Analysis (Peak Event)

- **Total Traffic:** 28M packets
- **Network Load:** 28M packets
- **Maximum Bytes Sent:** 35.285 KB per burst
- **Impact:** Abnormal network activity inconsistent with normal usage patterns, causing bandwidth saturation and potential service interruption.

Protocol Distribution

- TCP: 82.988K packets (2.04%)
- UDP: 555.04K packets (13.67%)
- ICMP: 1.43K packets (0.04%)
- **Implication:** Multi-protocol attack to evade simple filtering; requires deep packet inspection (DPI).

Targeted Victim Analysis

- **Victim IP:** 1.112.179.149
- **Observation:** 100% of high-volume traffic targeted this IP, confirming deliberate intent.
- **Business Impact:** Likely service degradation for users relying on the target server.

Top 10 Traffic Events

Rank	Total Traffic	Active Scanner	Target Victim	Network Load	Protocol Details
1	28M	147.32.84.59	1.112.179.149	28M	TCP 82.988K, UDP 555.04K, ICMP 1.43K
2	19M	147.32.84.191	1.112.179.149	19M	ICMP 294.43K
3	13M	84.202.57.30	1.112.179.149	13M	UDP 313.53K
4	12M	147.32.85.26	1.112.179.149	12M	TCP 218.95K, UDP 1.45K
5	10M	147.32.84.2	1.112.179.149	10M	TCP 184.57K
6	10M	213.89.44.168	1.112.179.149	10M	UDP 230.21K
7	9M	82.12.141.57	1.112.179.149	9M	UDP 214.05K
8	7M	147.32.84.165	1.112.179.149	7M	ICMP 98.49K
9	6M	88.100.184.137	1.112.179.149	7M	TCP 114.34K
10	5M	147.32.80.13	1.112.179.149	5M	TCP 96.22K, UDP 3.16K

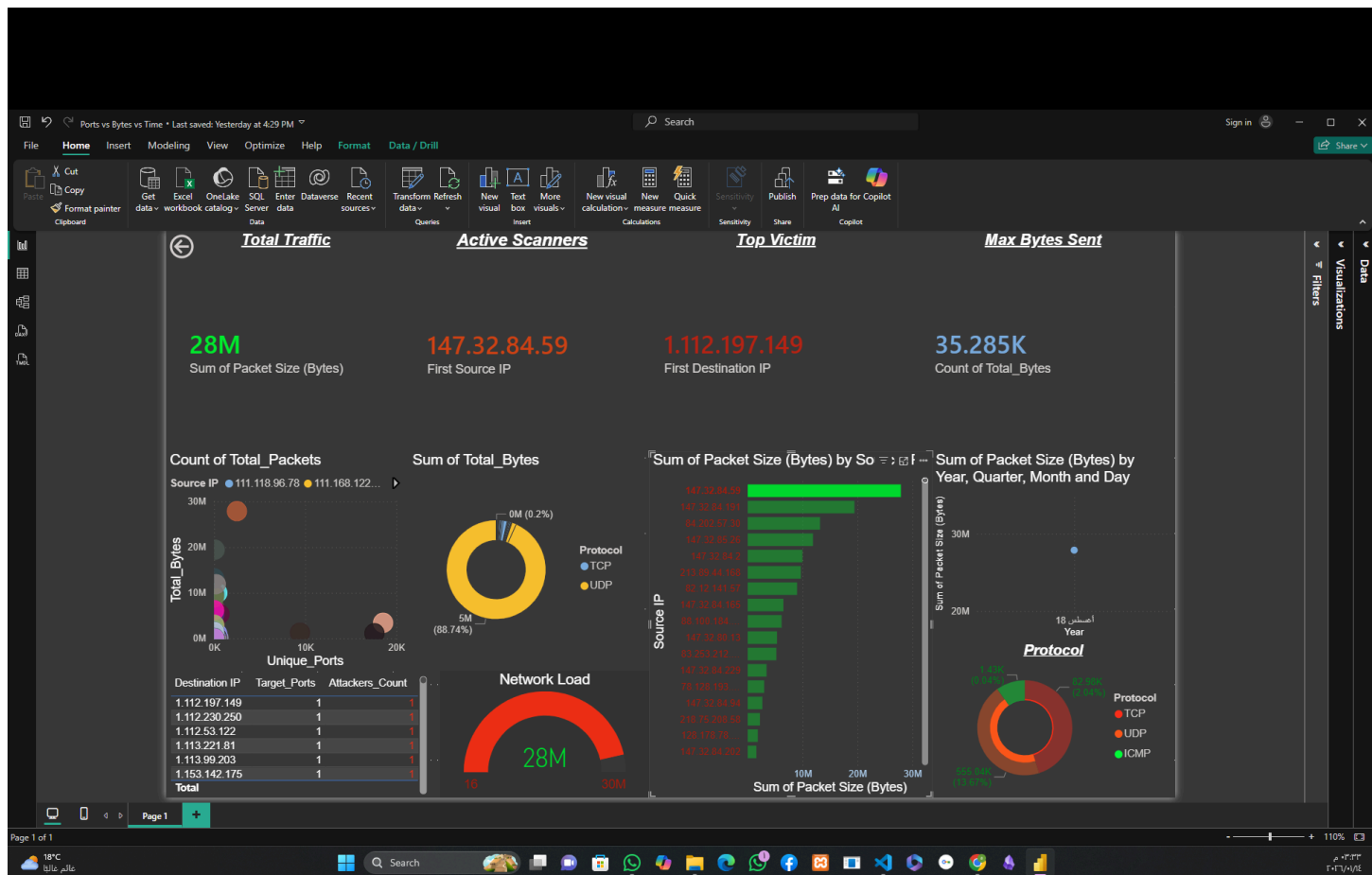
Indicators of Compromise (IOCs)

IOC Type	Value	Severity
Primary Source IP	147.32.84.59	Critical
Target Victim	1.112.179.149	High
Attack Vector	Volumetric Flood	High
Protocols	TCP, UDP, ICMP	Multi-vector

Power BI Dashboard Insights

The dashboard highlighted:

- Sudden traffic spikes correlating with Top 10 events
- TCP, UDP, and ICMP protocol distribution
- High-volume sources targeting a single victim



Conclusion

- Coordinated multi-protocol DDoS / Botnet attack confirmed.
- Peak traffic reached 28M packets from 147.32.84.59.
- Multiple high-volume sources targeting a single victim confirm Botnet behavior.
- Immediate mitigation required to restore network integrity.



Recommendations

1. **Block IPs:** Firewall block malicious source IPs listed in Top 10.
2. **Monitor:** Continuous monitoring for similar abnormal traffic patterns.
3. **Inspect Victim:** Forensic inspection of 1.112.179.149 for malware or compromise.
4. **Harden Network:** Apply rate limiting and protocol security measures to defend against volumetric attacks.



Technical Appendix (Full)

Analysis Tools & Libraries Used:

- **Python Libraries:**
 - pandas (Data manipulation & cleaning)
 - numpy (Numerical operations)
 - matplotlib.pyplot (Data plotting)
 - seaborn (Advanced statistical visualization)
 - plotly.express (Interactive visualizations)
 - datetime (Timestamp conversion)
 - ipaddress (IP address handling)
- **Visualization Setup:**

```
import matplotlib.pyplot as plt
import matplotlib.font_manager as fm
import seaborn as sns

sns.set(style="whitegrid")
plt.rcParams['figure.figsize'] = (12, 6)
%matplotlib inline
```